Chiron

Execution UI Consolidation and Platform Integration Under Controlled Scope

Progress Report 6 • 8 Dec 2025 — 21 Dec 2025

Master’s Research Progress Update

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Progress Report 6

Title: Execution UI Consolidation and Platform Integration Under Controlled Scope
Reporting Window: 8 Dec 2025 — 21 Dec 2025
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Abstract

Progress Report 6 was a consolidation-focused window with a deliberately controlled implementation scope. In product behavior, the period delivered targeted execution-surface improvements: a reusable workflow execution card and technique-flow corrections that improved completed-workflow usability. In parallel, the period absorbed large platform/documentation integration work, including external tooling content and research alignment for the next semester trajectory.

The key interpretation is balance: this was not a feature-expansion sprint in breadth, but a control sprint intended to reduce UX debt while preparing the project’s delivery environment for subsequent architecture and implementation phases. Commit cadence was compact, while touched-file volume was high due to integration-scale imports and repository reshaping.

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1. Period Character: Consolidation Before Expansion

Compared with the previous report, this window had a smaller number of product-facing development changes but very high integration/documentation surface movement. That pattern indicates intentional scope control:

- stabilize user-visible execution surfaces,
- avoid uncontrolled feature branching,
- and create cleaner platform/research foundations for the following phase.

This matches a consolidation objective rather than a capability race. Most repeated code hotspots remained concentrated in a narrow execution loop (workflow router/handler paths and related workflow UI components), while the largest file-volume movement came from documentation/tooling integration surfaces.

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2. Execution UI Consolidation

The primary user-facing addition was a reusable workflow execution card with progress visibility for steps/tools. This change matters because it improves consistency of execution-state presentation across workflow contexts and lowers UI duplication pressure.

In practical terms, this period moved execution visibility toward a more standardized component model, which is important for sustained supervision quality as workflow complexity grows.

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3. Technique Workflow Corrections and Completion Usability

Technique-path behavior received focused corrections, including stricter output behavior for at least one technique flow and clearer access to completed workflow views. These changes are modest in size but high in value for trust and interpretability:

- outputs became more aligned with expected technique semantics,
- and completed-run navigation improved user control over retrospective inspection.

This is consistent with the broader thesis direction: guided workflows must be inspectable both during execution and after completion.

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4. Large Integration Surface: External Platform Content and Repository Reshaping

A substantial part of the period was dominated by platform-level content movement and repository reshaping activity (including external integration imports and removal/replacement of prior bundled sources). This produced large touched-file volume, but the core narrative value is not raw volume; it is delivery-infrastructure readiness. The change pattern indicates migration/integration preparation rather than broad in-window runtime rewrites.

The period therefore combined two layers of work:

- **narrow product hardening** in execution surfaces,
- **broad integration groundwork** for tooling/research continuity.

This dual pattern explains why change-count alone under-represents period significance.

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5. Research and Workflow-Method Alignment

This window also strengthened research/process alignment through integration-oriented documentation work and workflow design notes intended to support upcoming implementation cycles. In this context, documentation should again be read as operational infrastructure rather than passive reporting.

The outcome is improved continuity between product execution behavior, methodology workflow definitions, and semester-level planning.

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6. Stack and Architecture Status at End of Progress Report 6

By the end of this period, the active direction remained within the same transition stack corridor established earlier:

- **Desktop host:** Tauri [3]
- **Frontend:** React + TanStack Router + shadcn/ui
- **Backend/API:** TypeScript services with Hono + tRPC [4], [5]
- **AI orchestration/runtime path:** Mastra + AX [1], [2]
- **Data layer:** Drizzle ORM with PostgreSQL
- **Authentication:** Better Auth
- **Runtime/tooling:** Bun + Turborepo

This report records no abrupt stack pivot inside the window. Instead, it documents consolidation and integration work that lowered operational risk before subsequent architecture-heavy changes.

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7. Why This Period Matters for the Next Phase

Progress Report 6 matters because it prevented execution and UX debt from compounding during a period of high integration churn. Without this stabilization window, later architecture decisions would have been evaluated on unstable product surfaces.

By combining targeted product corrections with large integration groundwork, the project entered the next phase with better execution consistency and clearer infrastructure boundaries.

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8. Open Questions Entering Progress Report 7

- Which parts of the new integration surface should be treated as stable platform dependencies versus experimental scaffolding?
- How should execution-card and completion-view patterns evolve as workflow states become more heterogeneous?
- What verification discipline is needed to ensure integration-scale changes do not regress workflow semantics?
- Which architecture decisions should be made next from direct runtime evidence versus planning assumptions?

These questions define the transition from consolidation toward the next layer of controlled expansion.

Conclusion

Progress Report 6 is best understood as a control sprint: limited but meaningful product-facing execution improvements combined with high-volume integration and repository-shaping work. The period intentionally favored stability, consistency, and infrastructure readiness over broad feature acceleration.

This framing keeps the narrative evidence-driven: what changed in user-visible execution was focused and concrete, while the broader project movement happened through platform integration and research-workflow alignment that prepared the next stage.

References

[1] Mastra, “Mastra Documentation,” 2025. [Online]. Available: https://mastra.ai/docs
[2] axllm, “ax,” GitHub repository (`@ax-llm/ax`), 2025. [Online]. Available: https://github.com/axllm/ax
[3] Tauri Programme, “Tauri,” Documentation, 2025. [Online]. Available: https://tauri.app/
[4] Hono Authors, “Hono,” Documentation, 2025. [Online]. Available: https://hono.dev/
[5] tRPC Authors, “tRPC,” Documentation, 2025. [Online]. Available: https://trpc.io/docs